

The Mechanism and Pathway of Asset Tokenization for Rural Revitalization: A Survey and Estimation Based on Xitan Village, Tianzhu County, Gansu Province

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Abstract: The idleness of rural assets and the lack of sustainability of fiscal investment are prominent contradictions in the process of rural revitalization. Based on an in-depth field survey in Xitan Village, Tianzhu County, Gansu Province, this paper proposes using Real-World Asset Tokenization (RWA) as a tool to activate dormant rural resources, achieving a transition from "project-based governance" to "cash-flow governance." The study finds that RWA can fundamentally restructure the cash-flow logic of rural asset operations by converting scattered land management rights, facility assets, and future income streams into measurable, auditable, tradable, and dividend-bearing digital certificates. Taking the construction of solar greenhouses on 2,500 mu of flat land in Xitan Village as the baseline, the first-year return on investment reaches 82.8%–105.6%, enabling a positive liquidity cycle of tokens within 11–14 months. If extended across Gansu Province, the total social capital investment required would be RMB 28.8 billion, with annual new revenues estimated at RMB 24.7–30.4 billion. The institutional value of RWA lies in its dual function: it is both a financial tool for leveraging social capital and a governance mechanism for safeguarding farmers' interests. This paper proposes an institutional framework of "farmers' interests first" and a three-stage implementation pathway (inventory, legislation, and pilot), providing a replicable model for revitalizing underdeveloped regions.

Keywords: Real-World Asset Tokenization (RWA); Rural Revitalization; Dormant Resources; Data Asset Governance; Cash-Flow Governance

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1 Introduction: The Problem Statement

As China enters a new phase of comprehensively promoting rural revitalization, rural development faces a fundamental dilemma. On the one hand, a large number of rural assets remain dormant—land lies idle, facilities are abandoned, and resources are inefficiently utilized. On the other hand, the fiscal-driven "blood-transfusion" model is unsustainable: after one-time investments, there is a lack of continuous operational and maintenance funds, and industrial projects often lack long-term income support after launch. How to shift from transfusion to hematopoiesis, and from project-driven to

cash-flow-driven development, has become a core issue in institutional innovation for rural revitalization.

Xitan Village in Tianzhu County, Gansu Province, is a typical case. Of the village's 7,737.74 mu of land, about 5,500 mu (72%) lies uncultivated. The government invested RMB 7.5 million to build 322.8 mu of solar greenhouses, but these are operated by a third-party enterprise. Villagers receive only RMB 360 per mu per year in land transfer fees, yielding a return rate of only 1.51%—merely 0.16 percentage points higher than the one-year fixed deposit rate. This pattern of "government builds, enterprises profit, farmers watch" reveals the deep-seated problems of farmers' absence from the

income stream and the lack of a cash-flow distribution mechanism in rural asset operations.

Real-World Asset Tokenization (RWA) offers a new solution to this dilemma. Using blockchain and smart contract technologies, RWA maps real-world assets—such as land management rights and agricultural product income rights—into on-chain digital certificates, turning dormant resources into divisible, tradable, and dividend-bearing digital assets. The core questions of this paper are: How can RWA activate dormant rural resources? What institutional logic should its design follow? And is it scalable in underdeveloped regions?

2 Theoretical Logic of RWA in Activating Dormant Rural Resources

2.1 The Transformation Chain: From Resources to Assets to Capital

The essence of dormant rural resources is "having resources but not assets." Land, facilities, and other physical entities exist, but they lack clear property rights definitions, stable cash-flow records, and tradable asset forms. RWA completes the transformation through three steps: First, titling and measurement—blockchain is used to record rights such as land management rights and income rights on-chain, forming immutable digital title certificates. Second, standardization and fractionalization—non-standardized physical assets are split into digital shares that can be held by different investors, lowering investment thresholds. Third, pricing and circulation—via trading platforms, price discovery and secondary-market circulation are achieved, forming a capital cycle of "investment – income – exit."

2.2 Cash-Flow Governance Logic to Break Fiscal Dependency

The traditional fiscal assistance model is essentially "project-based governance"—it targets project completion, uses fiscal appropriations as the primary means, and lacks institutional arrangements for the cash flow generated by asset operations. RWA, in contrast, builds a "cash-flow

governance" model: by linking future income (e.g., agricultural product sales) to digital certificates, it creates a predictable, distributable, and auditable cash-flow distribution mechanism. The key advantages of this mechanism are: it attracts social capital through market-based returns, automatically enforces distribution rules through smart contracts, and enables penetrating supervision via on-chain data, thereby overcoming the time-bound and finite nature of fiscal funds.

2.3 Synergy between RWA and Data Asset Governance

Agricultural data asset governance is the institutional foundation for RWA implementation. Data governance addresses the "asset pricing" issue—by establishing data asset systems such as soil moisture, yield quality, and market transaction data, it provides pricing references and risk-control parameters for RWA. RWA, in turn, solves the "data monetization" issue, converting the "data asset packages" formed by data governance into investable financial products. Together, they form a complete chain of "data collection → rights registration → assetization → tokenization → capital circulation," providing the institutional support for agricultural new-quality productive forces.

3 Benefit Estimation and Feasibility Analysis of the Xitan Village RWA Pilot

3.1 Village-Level Benefit Estimation

Based on field survey data from Xitan Village, if the 2,500 mu of idle flat land is used to build solar greenhouses, with each greenhouse occupying about 4 mu and costing about RMB 100,000, a total of 625 greenhouses can be built, with an initial investment of about RMB 62.5 million. Assuming oyster mushroom selling prices of RMB 2.5–3.0 per jin, production costs of RMB 1.2–1.4 per jin, and an annual yield of about 66,000 jin per greenhouse, each greenhouse generates annual profits of RMB 85,800–105,600. The total annual profit from 625 greenhouses reaches RMB 53.625–66 million,

yielding a first-year ROI of 82.8%–105.6%, allowing a positive token liquidity cycle within 11–14 months.

3.2 Expected Benefits of Provincial-Level Expansion

Using Xitan Village as the baseline, and drawing on Gansu's third national land survey data (78.1421 million mu of cultivated land, 9.6 million mu of uncultivated land, of which flat land accounts for 30%, and suitable uncultivated flat land for RWA development accounts for 40%), the total area available for RWA development in the province is 1.152 million mu. With an investment of RMB 25,000 per mu and an annual profit of RMB 21,400–26,400 per mu, the total investment demand is RMB 28.8 billion (led by social capital), and the expected annual new revenue is RMB 24.7–30.4 billion—about 7.7 times that of the traditional fiscal assistance model (which would yield less than RMB 3.6 billion annually for the same scale). Farmers' annual income increase is estimated at RMB 12.3–15.2 billion, and about 576,000 new local jobs would be created.

3.3 Feasibility Conditions

Gansu Province possesses the institutional soil for RWA pilots: digital infrastructure is steadily advancing, grassroots organizations and land-use control systems are relatively well-established, and the industrial scenarios in counties like Tianzhu are highly replicable. More importantly, the Xitan Village survey indicates that farmers are keenly aware of the low returns from land transfer and are hopeful that social capital entry can improve their incomes—this provides a social foundation for interest coordination and mass mobilization in the RWA pilot.

4 “Farmers’ Interests First”: The Institutional Architecture of RWA

To avoid repeating the trap of “capital goes to the countryside, farmers are sidelined,” RWA must translate the principle of “farmers’ interests first” from a value orientation into binding institutional constraints.

4.1 Institutionalized Distribution Order

Provincial-level management regulations should clearly stipulate: project revenues shall be prioritized for fulfilling guaranteed dividends and wage incomes to farmers and village collectives, secondarily for operation, maintenance, and reinvestment, and only lastly for distribution to external capital. Any adjustment to this distribution order must be reviewed and approved by three layers: the villagers’ representative assembly, the village collective, and the county-level supervisory task force, with public disclosure required.

4.2 Structural Arrangement of Governance Rights

A governance structure of “same shares, different rights” should be established: governance rights are primarily held by the village collective and farmers; external investment institutions are subject to position caps and voting rights caps (e.g., a single institution’s voting rights shall not exceed 30%). Major decisions must be approved by a “double majority” (more than half of farmers’ representatives and more than half of total voting rights), with a cooling-off period (e.g., a 72-hour reconsideration period) to prevent capital from using information asymmetry and voting advantages to force through resolutions that harm farmers’ interests at critical moments.

4.3 Traceability of Land-Use Control

All raised and operational funds must be tied to whitelisted accounts and usage tags, with on-chain fund flows and off-chain contracts and invoices being one-to-one matched to form a traceable closed loop. Any cross-category expenditure must undergo pre-authorization and post-audit procedures.

4.4 Phased Exit Arrangements

Phased exit and repurchase rules shall be set for three, five, and ten years to prevent concentrated social capital exits from disrupting the continuity of production and operations. Priority repurchase rights are granted to

the village collective and farmers, and repurchase prices are linked to asset appreciation.

5 Implementation Pathway: A Three-Stage Approach – Inventory, Legislation, Pilot

5.1 Stage One: Inventory and Survey to Solidify the Data Foundation

Taking the natural village as the basic unit, a three-month comprehensive survey shall be conducted to cover asset entities, sustainable cash flows, and industrial scenarios, resulting in three ledgers: the asset ledger (land management rights, facility assets, brands and geographical indications, etc.), the cash-flow ledger (detailed breakdown of costs and revenues over the past three years), and the constraint-boundary ledger (land-use controls, ecological red lines, water and energy consumption quotas, etc.). Data standards and coordinate systems shall be unified, and interfaces with real-estate registration, farmland protection monitoring, and other systems shall be established to ensure “data verified upon collection, traceable upon upload.”

5.2 Stage Two: Top-Level Design to Complete the Institutional Framework

Provincial authorities shall issue pilot management measures and supporting implementation rules, incorporating “farmers’ interests first, minimizing public risks, traceable land-use control, and non-transferable governance rights” as binding clauses. A white-paper template and three basic contract templates (dividend-based, mortgage-loan-based, and emergency-reserve-based) shall be published. A two-tier supervisory system consisting of county-level regulatory sandboxes and a provincial-level main node shall be established, with clear data disclosure standards and third-party auditing norms.

5.3 Stage Three: Pilot Projects to Generate Replicable Experience

Counties with sound digital infrastructure, clear industrial chains, and stable village-level governance

shall be selected as the first batch of pilots, with each county developing two to three differentiated scenario models. After one year of pilot operation, a third-party evaluation shall be conducted, and the mature experience shall be distilled into the County-Level Scenario Operation Manual, forming replicable institutional and technical assets.

6 Conclusions and Discussion

This study shows that RWA provides an efficient and equitable institutional tool for activating dormant rural resources. Its core value lies not in the technology itself, but in its ability to restructure the cash-flow logic and benefit-distribution mechanism of rural asset operations. Compared with the traditional fiscal “transfusion” model, RWA can leverage social capital, increase asset returns, and protect farmers’ interests. Compared with generic “capital-goes-to-the-countryside” models, RWA, through its binding “farmers’ interests first” institutional constraints and transparent on-chain supervision, can prevent capital from eroding farmers’ interests.

Nevertheless, promoting RWA in underdeveloped regions still faces numerous challenges: the urban-rural digital infrastructure gap, farmers’ limited digital literacy, lagging regulatory laws, and income volatility due to agricultural price fluctuations. Addressing these difficulties requires both technical solutions (e.g., simplified smart contracts, voice-interactive data collection) and institutional innovations (e.g., risk-compensation and insurance linkage mechanisms such as weather-index insurance and forward price-protection arrangements). Sustained farmer training and grassroots capacity building are also essential to solidify the social foundation for RWA’s healthy development. Future research may further investigate the adaptability of RWA to different agricultural scenarios, the integration of smart-contract governance with traditional rural governance, and the regulatory framework for cross-border RWA financing.

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